Exposition: A Stationary Gaussian Process Associated with the Critical-Line Zeta Function

Overview and Motivation

The paper constructs a stationary stochastic object from $\zeta\left(\frac{1}{2}+it\right)$, the Riemann zeta function restricted to the critical line. The central obstacle is that $\zeta\left(\frac{1}{2}+it\right)$ is not stationary in the usual sense: its zeros become denser and denser as $t \to \infty$, with the average gap between consecutive zeros near height T being approximately $2\pi/\log T$. A deterministic function with a time-varying zero density cannot have a well-defined, time-translation-invariant covariance. The paper resolves this by composing ζ with a time reparametrization that uniformizes the zero density, followed by a Jacobian normalization that preserves L^2 norm structure. The end product is a function $\zeta_{\mathrm{st}}(\tau)$ whose empirical covariance is exactly computable and identifies it as a sample path of a centered stationary Gaussian process with a specific rectangular spectral density.[1][2]

Part I: The Riemann–Siegel Theta Function

Definition and Role

The Riemann–Siegel theta function is defined for $t > 0$ by[3]

$$\vartheta(t) = \operatorname{Im}\log\Gamma\left(\tfrac{1}{4}+\tfrac{it}{2}\right) - \tfrac{t}{2}\log\pi.$$

Its primary significance is that it rotates $\zeta\left(\frac{1}{2}+it\right)$ by a real phase, turning it into the Hardy Z-function $Z(t) = e^{i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right)$, which is real-valued on the real axis. The function ϑ appears directly in the

zero-counting formula. The Riemann–von Mangoldt formula states[4][5][1]

$$N(T) = \frac{\vartheta(T)}{\pi} + 1 + S(T), \qquad S(T) = O(\log T),$$

so $\vartheta(T)/\pi$ accounts for essentially all the zero count up to height T. This makes $\tau = \vartheta(t)$ the natural "uniformized time" variable: in the τ-coordinate, zeros accumulate at a constant average rate.

Derivative Formulas and Polygamma Functions

The paper derives ϑ' by differentiating through the log-gamma function. Setting $z(t) = \frac{1}{4} + \frac{it}{2}$, one differentiates $\log\Gamma(z(t))$ to get $\psi(z(t)) \cdot \frac{i}{2}$, where $\psi = (\log\Gamma)'$ is the digamma function. The identity $\mathrm{Im}(iw) = \mathrm{Re}(w)$ for any complex w immediately gives

$$\vartheta'(t) = \tfrac{1}{2}\mathrm{Re}\ \psi\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \tfrac{1}{2}\log\pi.$$

Higher derivatives follow analogously, with the n-th derivative of $\log\Gamma(z(t))$ being $\psi^{(n-1)}(z(t)) \cdot (i/2)^n$; taking imaginary parts converts this to the real formula stated.

Stirling Asymptotics

Applying Stirling's formula $\log\Gamma(z) = (z - \frac{1}{2})\log z - z + \frac{1}{2}\log(2\pi) + O(z^{-1})$ to $z = \frac{1}{4} + \frac{it}{2}$ and extracting the imaginary part yields the asymptotic[3]

$$\vartheta(t) = \tfrac{t}{2}\log\tfrac{t}{2\pi e} - \tfrac{\pi}{8} + O(t^{-1}).$$

Differentiating this asymptotic gives

$$\vartheta'(t) = \tfrac{1}{2}\log\tfrac{t}{2\pi} + O(t^{-2}).$$

Since $\frac{1}{2}\log(t/2\pi) \to +\infty$ as $t \to \infty$, the derivative is eventually positive and diverges. This is the key fact that makes ϑ eventually strictly increasing.

Existence of the Monotone Branch

The function ϑ is not globally monotone on $(0,\infty)$: it decreases on an initial compact segment and has local extrema near $t \approx \pm 6.29$, reaching values approximately ∓ 3.53. The paper defines T_0 as the supremum of $\{t > 0 : \vartheta'(t) \le 0\}$. Since $\vartheta'(t) > 0$ for all sufficiently large t, this set is bounded above, so $T_0 < \infty$. By continuity, $\vartheta'(T_0) = 0$ and $\vartheta'(t) > 0$ for all $t > T_0$. This gives a globally defined, strictly increasing branch[3]

$$\vartheta \restriction_{[T_0,\infty)} \colon [T_0,\infty) \to [\tau_0,\infty), \qquad \tau_0 = \vartheta(T_0).$$

The exact numerical value of T_0 is irrelevant to all asymptotic results; it contributes only $O(1)$ to quantities that diverge.

Lagrange Inversion of ϑ^{-1}

Because ϑ is real-analytic and $\vartheta'(t_0) > 0$ at any $t_0 \ge T_0$, the inverse function theorem guarantees a real-analytic local inverse. The paper makes this explicit via the Lagrange inversion theorem, which gives the power series expansion of the inverse of an analytic function F near a simple zero of F. Setting $F(t) = \vartheta(t) - \tau_0'$ with $\tau_0' = \vartheta(t_0)$, the theorem produces[6][7]

$$\vartheta^{-1}(\tau) = t_0 + \sum_{k=1}^{\infty} \frac{c_k}{k!}(\tau - \tau_0')^k, \qquad c_k = \frac{d^{k-1}}{dt^{k-1}} \left[\frac{1}{\vartheta'(t)}\right]^k \Bigg|_{t=t_0}.$$

Each coefficient c_k is a polynomial in the values $\{\vartheta^{(j)}(t_0) : 1 \le j \le k\}$, which are computable from the polygamma formulas. The series converges on any interval $(\tau_0', \vartheta(t_0+\varepsilon))$ where $\varepsilon < \inf_{t \ge T_0} 1/|\vartheta''(t)/\vartheta'(t)^2|$.

Part II: The Stationary Zeta Function

Definition and the Two Normalization Requirements

For $\tau > \tau_0$, the paper defines

$$\zeta_{\mathrm{st}}(\tau) = \frac{\zeta\big(\frac{1}{2} + i\,\vartheta^{-1}(\tau)\big)}{\sqrt{2\,\vartheta'(\vartheta^{-1}(\tau))}}.$$

The denominator serves two simultaneous purposes that together fix it uniquely.

The first purpose is **zero uniformization**. In the τ-variable, the zeros of ζ_{st} are precisely the images $\vartheta(t_\rho)$ of the zeros t_ρ of ζ on the critical line. The zero-counting result then gives

$$\frac{\#\{\tau \in (\tau_0, T) : \zeta_{\mathrm{st}}(\tau) = 0\}}{T} = \frac{N(\vartheta^{-1}(T))}{T} = \frac{T/\pi + 1 + O(\log}{T}$$

The denominator $\sqrt{\vartheta'(\vartheta^{-1}(\tau))}$ is the Jacobian of the change of variables $\tau = \vartheta(t)$, and it is precisely this change of variables that converts the varying zero density $\vartheta'(t)/\pi \sim \frac{1}{2\pi}\log\frac{t}{2\pi}$ into the constant rate $1/\pi$.[1]

The second purpose is the **unitary time change**. The substitution $\tau = \vartheta(t)$ satisfies

$$\int_{\tau_0}^{\infty} \frac{|f(\vartheta^{-1}(\tau))|^2}{\vartheta'(\vartheta^{-1}(\tau))}\,d\tau = \int_{T_0}^{\infty} |f(t)|^2\,dt,$$

which means the map $f \mapsto f \circ \vartheta^{-1}/\sqrt{\vartheta' \circ \vartheta^{-1}}$ is unitary between $L^2(T_0, \infty)$ and $L^2(\tau_0, \infty)$. This is the standard L^2 change-of-variables isometry. The factor of 2 is then fixed by requiring the Cesàro mean square to equal 1.[8]

Unit Second Moment

The paper establishes that the long-run mean square of ζ_{st} equals 1. Substituting $\tau = \vartheta(t)$, the Jacobian factor cancels:

$$\frac{1}{T - \tau_0}\int_{\tau_0}^{T} |\zeta_{\mathrm{st}}(\tau)|^2\,d\tau = \frac{1}{2(\vartheta(S) - \tau_0)}\int_{T_0}^{S} \left|\zeta\big(\tfrac{1}{2} + it\big)\right|^2 dt,$$

with $T = \vartheta(S)$. The Hardy–Littlewood mean-square formula gives[9][10]

$$\int_0^S \left|\zeta\left(\tfrac{1}{2}+it\right)\right|^2 dt \sim S\log S.$$

Combined with $\vartheta(S) \sim \frac{S}{2}\log S$ from the Stirling asymptotics, the ratio is

$$\frac{S\log S}{2\cdot\frac{S}{2}\log S} = 1.$$

The uniqueness of the normalization follows from the same calculation: any positive measurable $e(\tau)$ producing unit mean square must satisfy $e(\tau)^2 \sim 2\vartheta'(\vartheta^{-1}(\tau))$ as $\tau \to \infty$.

Part III: The Hardy Z-Function Representation

The Hardy Z-function[4]

$$Z(t) = e^{i\vartheta(t)}\zeta\left(\tfrac{1}{2}+it\right)$$

is real-valued for real t, which is immediate from the functional equation of ζ: conjugating $\zeta\left(\frac{1}{2}+it\right)$ produces $e^{-2i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right)$, so multiplying by $e^{i\vartheta(t)}$ gives a real number. Since $\vartheta(\vartheta^{-1}(\tau)) = \tau$, the stationary function takes the form

$$\zeta_{\text{st}}(\tau) = e^{-i\tau}\cdot\frac{Z(\vartheta^{-1}(\tau))}{\sqrt{2\,\vartheta'(\vartheta^{-1}(\tau))}}.$$

This decomposition separates ζ_{st} into a pure oscillatory factor $e^{-i\tau}$ and a real amplitude $Z(\vartheta^{-1}(\tau))/\sqrt{2\vartheta'(\vartheta^{-1}(\tau))}$. The pure phase $e^{-i\tau}$ is responsible for the shift in the final spectral density from the symmetric interval $[-1,1]$ to the interval $[-2,0]$.

Part IV: Autocovariance Computation

Reduction to a ZZ-Average

The empirical covariance is defined as the Cesàro limit

$$R(\Delta) = \lim_{T\to\infty} \frac{1}{T-\tau_0} \int_{\tau_0}^{T} \zeta_{\mathrm{st}}(\tau+\Delta)\,\overline{\zeta_{\mathrm{st}}(\tau)}\,d\tau.$$

After substituting $\tau = \vartheta(t)$ and using the Z-function representation, the integrand factors into a phase $e^{-i\Delta}$, an amplitude ratio involving $\vartheta'(s)/\vartheta'(t)$ where $s = \vartheta^{-1}(\vartheta(t)+\Delta)$, and the product $Z(s)Z(t)$. The key asymptotic estimate is that

$$\frac{\vartheta'(s)}{\vartheta'(t)} = 1 + O\left(\frac{\Delta}{t(\log t)^2}\right),$$

which follows from $\vartheta'(t) \sim \frac{1}{2}\log(t/2\pi)$ and $\vartheta''(t) = O(t^{-1})$, together with the Lagrange expansion $s(t,\Delta) = t + \Delta/\vartheta'(t) + O(\Delta^2\vartheta''(t)/\vartheta'(t)^3)$. After averaging, this error is $O((\log S)^{-1}) \to 0$, leaving

$$R(\Delta) = e^{-i\Delta} \lim_{S\to\infty} \frac{1}{2(\vartheta(S)-\tau_0)} \int_{T_0}^{S} Z(s(t,\Delta))\,Z(t)\,dt.$$

The Riemann–Siegel Expansion

The paper uses the Riemann–Siegel formula[4]

$$Z(t) = 2 \sum_{n\le\sqrt{t/(2\pi)}} \frac{\cos(\vartheta(t) - t\log n)}{\sqrt{n}} + O(t^{-1/4}),$$

which follows from the approximate functional equation for ζ by multiplying both Dirichlet sums by $e^{i\vartheta(t)}$ and using the functional equation to combine them. Substituting into the ZZ-product expands the integrand as a double sum over pairs (m,n). The product-to-sum identity $2\cos A\cos B = \cos(A-B) + \cos(A+B)$ splits each term into a "minus part" with phase $t\log(n/m) + \alpha_m(t)$ and a "plus part" with phase $2\vartheta(t) - t\log(mn) + \alpha_m(t)$.

Elimination of Non-Diagonal Terms

The $(+)$-part vanishes. The phase $2\vartheta(t) - t\log(mn) + \alpha_m(t)$ has derivative $2\vartheta'(t) - \log(mn) + O(1/t\log^2 t)$ with respect to t. Since $\log(mn) \le 2\log M(t) \sim \log(t/2\pi) \sim 2\vartheta'(t)$, the derivative is bounded away from zero or grows, so integration by parts bounds each fixed-pair integral by $O(1)$. Summing over all $O(S)$ pairs and dividing by $\vartheta(S) \sim \frac{S}{2}\log S$ gives $O((\log S)^{-1}) \to 0$.

The off-diagonal $(-)$-part vanishes. For $m \ne n$, the phase derivative is $\log(n/m) + O(\Delta/t\log^2 t) \asymp k/m$ where $k = |n-m| \ge 1$. Integration by parts bounds the integral for each pair by $O(m/k)$. Using $1/\sqrt{mn} \le 1/m$, each term is $O(1/k)$. Summing over all pairs with offset k gives a total of $O(M(S)\log M(S)) \asymp O(\sqrt{S}\log S)$, and dividing by $\vartheta(S) \asymp S\log S$ gives $O(S^{-1/2}) \to 0$.

The Diagonal Calculation

On the diagonal $m = n$, the phase simplifies to

$$\alpha_n(t) = \Delta\left(1 - \frac{\log n}{\vartheta'(t)}\right).$$

The variable $u_n(t) = \log n/\vartheta'(t)$ ranges over $[0, 1+o(1)]$ for $n \le M(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$, since $\log M(t) \approx \vartheta'(t)$. Applying the Euler–Maclaurin formula to convert the sum over n to an integral converts the diagonal sum to[11]

$$\sum_{n\le M(t)} \frac{\cos(\Delta(1-u_n(t)))}{n} = \vartheta'(t)\int_0^1 \cos(\Delta(1-u))\,du + o(\vartheta'(t)$$

The integral $\int_{T_0}^S \vartheta'(t)\,dt = \vartheta(S) - \tau_0$ cancels the denominator, leaving

$$2\int_0^1 \cos(\Delta(1-u))\,du = 2\int_0^1 \cos(\Delta v)\,dv = \frac{2\sin\Delta}{\Delta}.$$

The prefactor of $\frac{1}{2}$ from the product-to-sum identity reduces this to $\sin\Delta/\Delta$, and multiplying by $e^{-i\Delta}$ from the reduction lemma gives the final result

$$R(\Delta) = e^{-i\Delta}\frac{\sin\Delta}{\Delta}, \qquad R(0) = 1.$$

Part V: Spectral Density and the Gaussian Process

Fourier Transform and the Rectangular Spectrum

The Fourier transform of $\sin\Delta/\Delta$ is a classical result:[12][13]

$$\widehat{\left(\frac{\sin\Delta}{\Delta}\right)}(\omega) = \pi\,\mathbf{1}_{[-1,1]}(\omega).$$

Multiplication of $f(\Delta)$ by $e^{-i\Delta}$ shifts the Fourier transform: $\widehat{e^{-i\Delta}f}(\omega) = \widehat{f}(\omega+1)$. Applied to $f(\Delta) = \sin\Delta/\Delta$:

$$\widehat{R}(\omega) = \pi\,\mathbf{1}_{[-1,1]}(\omega+1) = \pi\,\mathbf{1}_{[-2,0]}(\omega).$$

The spectral support $[-2,0]$ arises precisely from the phase $e^{-i\Delta}$ in the covariance, which shifts the symmetric sinc spectrum by -1. This means ζ_{st} has bandwidth 2 and is a bandpass signal centered at frequency -1.

Bochner's Theorem and Existence of the Gaussian Process

Bochner's theorem states that a continuous function $R:\mathbb{R}\to\mathbb{C}$ is the covariance of a weakly stationary process if and only if its Fourier transform is a non-negative measure. Since $\widehat{R}(\omega) = \pi\,\mathbf{1}_{[-2,0]}(\omega) \geq 0$, the covariance $R(\Delta) = e^{-i\Delta}\sin\Delta/\Delta$ is positive definite. The Kolmogorov existence theorem then guarantees a centered stationary Gaussian process $(X(\tau))_{\tau\in\mathbb{R}}$ with[14][15][16][17]

$$\mathbb{E}\Big[X(\tau+\Delta)\,\overline{X(\tau)}\Big] = e^{-i\Delta}\,\frac{\sin\Delta}{\Delta}$$

for all $\tau, \Delta \in \mathbb{R}$. The process is identified at the level of empirical second-order statistics: the Cesàro averages of ζ_{st} reproduce exactly the covariance of the Gaussian process.

Mathematical Architecture: Key Dependencies

The argument has a clean layered structure:

Layer	Content	Tools used
1. Monotonicity of ϑ	T_0 exists and is finite	Stirling asymptotics of ϑ'
2. Inversion	ϑ^{-1} exists, is real-analytic, Lagrange series	Inverse function theorem, Lagrange inversion
3. Normalization	Factor $\sqrt{2\vartheta'(\vartheta^{-1}(\tau))}$ is unique	Hardy–Littlewood mean square, $\vartheta(S) \sim \frac{S}{2}\log S$
4. Zero rate	Rate $1/\pi$ in τ-variable	Riemann–von Mangoldt formula
5. Covariance	$R(\Delta) = e^{-i\Delta}\sin\Delta/\Delta$	Riemann–Siegel formula, Euler–Maclaurin, integration by parts
6. Process	Stationary Gaussian process exists	Bochner's theorem, Kolmogorov existence

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Remarks on Hypotheses and Scope

No Riemann Hypothesis assumed. All results are unconditional. The zero-counting formula $N(T) = \vartheta(T)/\pi + 1 + S(T)$ with $S(T) = O(\log T)$ is established unconditionally by the Riemann–von Mangoldt formula[1][5]. The Hardy–Littlewood mean-square estimate $\int_0^T |\zeta(\frac{1}{2} + it)|^2\, dt \sim T\log T$ is also unconditional[9][10]. The only input about zeros is their counting function, not their locations.

Connection to random matrix theory. The identification of ζ_{st} as a sample path of a specific Gaussian process with rectangular spectral density connects to the broader program of modeling the zeta function by random matrix ensembles. The unitary group $U(N)$ has eigenvalue density that becomes uniform on the circle as $N \to \infty$, and the reparametrization $\tau = \vartheta(t)$ is the arithmetic analogue of this uniformization. The rectangular spectral density with support $[-2, 0]$ and the unit bandwidth characteristic are intrinsic properties of the model, not artifacts of approximation.[18][19][20][21]

Empirical versus distributional covariance. The paper carefully distinguishes the empirical covariance—a Cesàro limit of a deterministic integral—from the distributional covariance of a genuine random process. The identification is at the level of second-order statistics only: ζ_{st} is a deterministic function whose two-point Cesàro averages match the covariance of the Gaussian process. This does not assert that ζ_{st} is a Gaussian process or that its higher-order statistics match the Gaussian model.

The factor of 2. The denominator $\sqrt{2\vartheta'(\vartheta^{-1}(\tau))}$ rather than just $\sqrt{\vartheta'(\vartheta^{-1}(\tau))}$ is forced by the asymptotic $\int_0^S |\zeta(\frac{1}{2} + it)|^2\, dt \sim S \log S$ together with $\vartheta(S) \sim \frac{S}{2} \log S$. Without the factor of 2 in the denominator, the Cesàro mean square would be 2 rather than 1, and the covariance would be $R(0) = 2$ rather than 1. The uniqueness theorem in the paper shows this is the only normalization achieving $R(0) = 1$ under the given asymptotic data.

References

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